

Gen-Z Social Media Usage Analysis Dashboard

1. Introduction

The **Gen-Z Social Media Usage Analysis Dashboard** is a comprehensive end-to-end analytics project developed to study social media engagement patterns, behavioral trends, addiction levels, and mental health insights among Gen-Z users. The project focuses on understanding how digital habits and social media consumption influence emotional wellness and user behavior.

This project was developed using a complete analytics workflow that includes:

- Data collection and preprocessing using Microsoft Excel
- Data storage and querying using PostgreSQL
- Data analysis and transformation using Python
- Interactive dashboard development using Power BI
- DAX calculations and KPI-based analytical reporting

The dashboard was designed with a modern analytics theme and provides visually rich insights through multiple interactive pages.

2. Objectives

The primary objectives of the project are:

- To analyze social media usage patterns among Gen-Z users
- To study the relationship between screen time and mental health
- To identify addiction behavior and platform engagement trends
- To create interactive dashboards for analytical storytelling
- To transform raw social media data into actionable insights
- To improve practical skills in data analytics and business intelligence tools

3. Technologies and Tools Used

Technology	Purpose
Microsoft Excel	Data cleaning and preprocessing
Python	Data analysis and transformation
PostgreSQL	Database storage and SQL querying
Power BI	Dashboard visualization and reporting
DAX	KPI calculations and custom measures

4. Excel Data Preprocessing

Microsoft Excel was used for initial preprocessing and preparation of the dataset before importing it into PostgreSQL and Power BI.

Activities Performed in Excel

- Removed duplicate records
- Checked missing and null values
- Standardized column names
- Corrected inconsistent formatting
- Organized categorical values
- Performed basic filtering and sorting
- Verified data quality and integrity

Benefits of Excel in the Project

Excel helped in preparing a clean and structured dataset suitable for database storage and dashboard analysis. It simplified initial inspection of user behavior records and ensured the dataset was ready for further analytical processing.

cleaned_genz_social_media														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	age	gender	country	daily_usage_hours	primary_platform	num_platforms_used	purpose	avg_session_minutes	night_usage	mental_health_score	addiction_level	screen_time_before_sleep	mental_health_status	daily_usage_status
1	19	Male	Australia	2.66	Snapchat	4	Education	29.22	0	6.71	Medium	15.8	Average	Moderate Usage
2	16	Female	USA	5.69	Twitter	3	Socializing	32.46	1	6.63	High	47.7	Average	High Usage
3	25	Female	India	2.78	TikTok	2	Socializing	18.17	1	6.04	Medium	62.5	Average	Moderate Usage
4	27	Male	India	3.49	YouTube	4	Education	22.49	1	8.11	Medium	31.9	Good	Moderate Usage
5	23	Female	India	3.75	TikTok	1	Entertainment	27.36	1	6.66	Medium	29.5	Average	Moderate Usage
6	20	Other	Brazil	2.93	TikTok	1	Entertainment	37.43	1	7.14	Medium	55.2	Good	Moderate Usage
7	25	Male	Germany	5.24	TikTok	1	Entertainment	31.18	1	6.53	High	71.7	Average	High Usage
8	17	Female	India	2.24	YouTube	1	News	21.72	0	9.31	Low	12.3	Good	Moderate Usage
9	19	Female	Brazil	2.78	Twitter	2	Socializing	27.58	1	6.3	Medium	43.3	Average	Moderate Usage
10	22	Male	Canada	3.59	Instagram	1	Education	5	0	6.68	Medium	42.9	Average	Moderate Usage
11	15	Female	USA	1.74	Twitter	2	Entertainment	7.06	1	8.16	Low	30.7	Good	Low Usage
12	19	Male	India	3.73	Twitter	2	Education	5	1	7.91	Medium	8.6	Good	Moderate Usage
13	23	Female	USA	6.21	Instagram	1	Socializing	11.25	1	5.64	High	53.8	Average	Moderate Usage
14	23	Female	Australia	3.74	TikTok	1	News	24.21	1	6.31	Medium	44	Average	Moderate Usage
15	20	Male	India	5.16	Instagram	5	Education	20.29	1	4.56	High	7.2	Bad	High Usage
16	17	Male	UK	4.37	YouTube	1	Entertainment	28.61	0	7.24	Medium	0	Good	Moderate Usage
17	16	Male	Australia	5.33	TikTok	1	Entertainment	13.05	1	6.11	High	49.8	Average	High Usage
18	20	Male	USA	3.37	Twitter	3	Socializing	20.44	1	6.31	Medium	6.2	Average	Moderate Usage
19	20	Male	USA	3.9	Snapchat	2	Entertainment	14.4	1	8.59	Medium	15.5	Good	Moderate Usage
20	15	Female	USA	4.89	Instagram	2	Education	31.86	1	6.48	Medium	21.9	Average	Moderate Usage
21	18	Female	Australia	2.5	YouTube	3	Entertainment	29.22	0	9	Medium	39.6	Good	Moderate Usage
22	17	Male	UK	5.12	YouTube	1	Entertainment	14.77	1	6.19	High	49.6	Average	High Usage
23	14	Male	Canada	3.69	Snapchat	4	Entertainment	21.37	0	6.26	Medium	24.1	Average	Moderate Usage
24	25	Male	India	1.94	Instagram	5	Education	11.66	1	7.52	Low	53.7	Good	Low Usage
25	24	Male	USA	2.67	YouTube	5	Socializing	38.05	1	6.74	Medium	22.9	Average	Moderate Usage
26	26	Male	Australia	5.6	Instagram	3	Entertainment	40.19	1	4.33	High	52.9	Bad	High Usage
27	18	Male	Brazil	1.2	Instagram	4	Entertainment	37.3	0	8.93	Low	46.7	Good	Low Usage

cleaned_genz_social_media														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	age	gender	country	daily_usage_hours	primary_platform	num_platforms_used	purpose	avg_session_minutes	night_usage	mental_health_score	addiction_level	screen_time_before_sleep	mental_health_status	daily_usage_status
999975	16	Male	India	3.81	Twitter	2	News	28.16	1	6.59	Medium	21.1	Average	Moderate Usage
999976	20	Female	USA	4.03	Twitter	1	Socializing	28.63	1	7.37	Medium	36.4	Average	Moderate Usage
999977	26	Female	India	3.48	TikTok	4	Entertainment	32	1	8.53	Medium	41.5	Good	Moderate Usage
999978	24	Other	USA	3.05	TikTok	3	News	22.5	0	7.2	Medium	79.7	Average	Moderate Usage
999979	15	Female	Germany	5.26	Instagram	3	Socializing	25.22	0	4.61	High	24.7	Good	High Usage
999980	19	Male	India	6.32	TikTok	3	Socializing	23.79	1	4.92	High	34.5	Average	High Usage
999981	17	Female	USA	3.98	Instagram	3	News	12.64	1	7.23	Medium	46.9	Good	Moderate Usage
999982	24	Male	India	1.38	YouTube	5	Socializing	58.46	0	8.21	Low	54.2	Good	Low Usage
999983	13	Female	USA	4.03	YouTube	2	Socializing	27.34	0	5.57	Medium	51.2	Good	Moderate Usage
999984	23	Female	USA	2.78	YouTube	2	News	15.14	0	7.7	Medium	34.8	Good	Moderate Usage
999985	26	Female	Brazil	4.23	Twitter	1	Entertainment	19.88	1	6.6	Medium	47.8	Good	Moderate Usage
999986	24	Male	Canada	0.54	YouTube	5	Education	22.82	1	9.73	Low	46.6	Good	Low Usage
999987	23	Female	UK	5.62	YouTube	5	Socializing	21.29	1	3.81	High	15.4	Average	High Usage
999988	23	Female	UK	2.48	TikTok	2	Entertainment	27.18	1	8.13	Low	53.3	Average	Moderate Usage
999989	17	Female	Canada	2.75	TikTok	5	Entertainment	49.13	1	10	Medium	65.2	Bad	Moderate Usage
999990	21	Female	India	1.71	Twitter	2	Education	7.44	1	8.71	Low	72.2	Average	Low Usage
999991	25	Male	USA	5.18	YouTube	5	Entertainment	19.89	1	7.23	High	21.9	Good	High Usage
999992	15	Female	India	2.01	YouTube	5	Socializing	26.05	1	7.62	Low	19.9	Average	Moderate Usage
999993	24	Female	UK	5.52	TikTok	4	Education	14.19	1	5.27	High	48.9	Good	High Usage
999994	22	Female	Germany	3.49	Snapchat	3	Education	28.1	1	6.73	Medium	38.3	Good	Moderate Usage
999995	27	Male	UK	5.24	Twitter	4	Education	5.27	1	5.59	High	51.3	Good	High Usage
999996	14	Other	USA	2.57	Snapchat	1	Entertainment	27.43	1	6.77	Medium	26.2	Good	Moderate Usage
999997	21	Male	India	3.1	Snapchat	4	Entertainment	21.16	1	7.23	Medium	66.3	Good	Moderate Usage
999998	14	Male	India	3.66	Instagram	1	Education	26.31	1	8.27	Medium	47.2	Good	Moderate Usage
999999	20	Male	India	4.47	YouTube	2	Socializing	13.62	0	7.61	Medium	18.9	Average	Moderate Usage
1000000	24	Female	Germany	7.83	YouTube	2	Content Creation	15.79	1	2.34	High	62.1	Bad	High Usage
1000001	13	Male	Canada	5.31	TikTok	1	Socializing	26.4	1	5.83	High	34.9	Average	High Usage

5. Python Data Analysis

Python was used for data analysis and exploratory analysis before visualization. Libraries such as Pandas and NumPy were used to understand behavioral patterns and relationships between variables.

Python Activities Performed

- Dataset loading and inspection
- Exploratory Data Analysis (EDA)
- Correlation analysis
- Statistical calculations
- Aggregation and grouping operations
- Mental health trend analysis
- Screen time behavior analysis

Python Libraries Used

- Pandas
- NumPy
- Matplotlib
- Seaborn

Sample Analytical Insights

- Users with higher screen time showed lower mental health scores
- Instagram and TikTok showed the highest engagement among Gen-Z users
- Medium addiction users formed the largest user segment
- Working Gen-Z and Young Adults showed higher platform engagement

Python analysis helped identify important behavioral trends that were later visualized in Power BI dashboards.

6. PostgreSQL Database Management

PostgreSQL was used to manage and organize the dataset in a structured relational database environment.

Database Activities

- Imported cleaned dataset into PostgreSQL
- Created tables for structured storage
- Performed SQL queries for filtering and aggregation
- Validated dataset consistency
- Retrieved analytical records for Power BI integration

SQL Operations Used

- SELECT statements
- GROUP BY operations
- Aggregate functions
- Filtering conditions
- Sorting and data retrieval

Benefits of PostgreSQL

Using PostgreSQL improved data handling efficiency and provided real-world experience in database management and SQL-based analytics workflows.

jupyter

GenZ_SocialMedia_Analysis

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Code

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JupyterLabPython [conda env:base]Anaconda Toolbox

[15]:import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[16]:df = pd.read_csv("cleaned_genz_social_media.csv")

[17]:df.head()

[17]:

	age	gender	country	daily_usage_hours	primary_platform	num_platforms_used	purpose	avg_session_minutes	night_usage	mental_health_score	addiction
0	19	Male	Australia	2.66	Snapchat	4	Education	29.22	0	6.71	M
1	16	Female	USA	5.69	Twitter	3	Socializing	32.46	1	6.63	
2	25	Female	India	2.78	TikTok	2	Socializing	18.17	1	6.04	M
3	27	Male	India	3.49	YouTube	4	Education	22.49	1	8.11	M
4	23	Female	India	3.75	TikTok	1	Entertainment	27.36	1	6.66	M

[11]:df.shape

[11]:(1000000, 15)

```
[29]: df.isnull().sum()
```

```
[29]: age                0
gender              0
country            0
daily_usage_hours  0
primary_platform   0
num_platforms_used 0
purpose            0
avg_session_minutes 0
night_usage        0
mental_health_score 0
addiction_level    0
screen_time_before_sleep 0
mental_health_status 0
daily_usage_status 0
age_group          0
dtype: int64
```

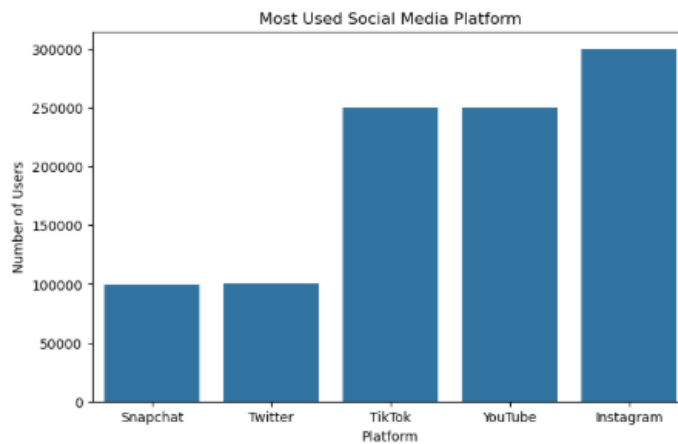
```
[30]: df['daily_usage_status'].value_counts()
```

```
[30]: daily_usage_status
Moderate Usage    684467
High Usage        158130
Low Usage         157399
Name: count, dtype: int64
```

```
[31]: df['age_group'].value_counts()
```

```
[31]: age_group
Working Gen-Z    333818
Young Adults     266116
Mid Teens        280455
Early Teens      280487
Name: count, dtype: int64
```

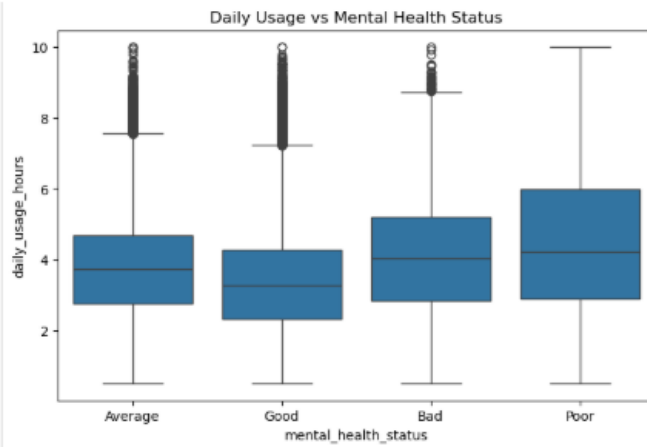
```
[19]: plt.figure(figsize=(8,5))
sns.countplot(x='primary_platform',data=df)
plt.title("Most Used Social Media Platform")
plt.xlabel("Platform")
plt.ylabel("Number of Users")
plt.show()
```



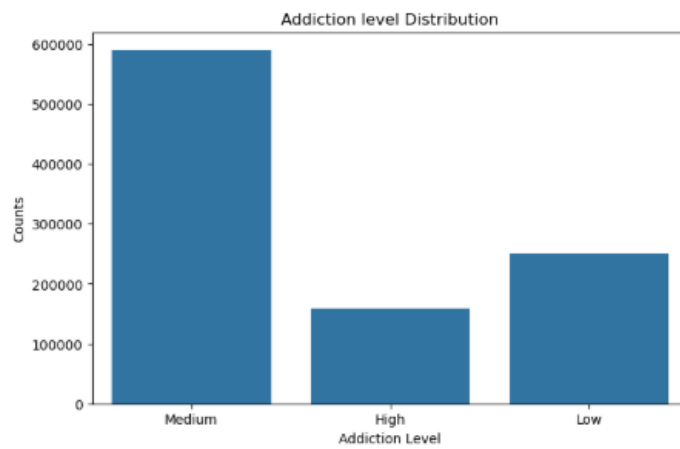
```
[21]: plt.figure(figsize=(8,5))
```

Mental Health Status

```
[22]: plt.figure(figsize=(8,5))
sns.boxplot(x='mental_health_status',y='daily_usage_hours',data=df)
plt.title("Daily Usage vs Mental Health Status")
plt.show()
```



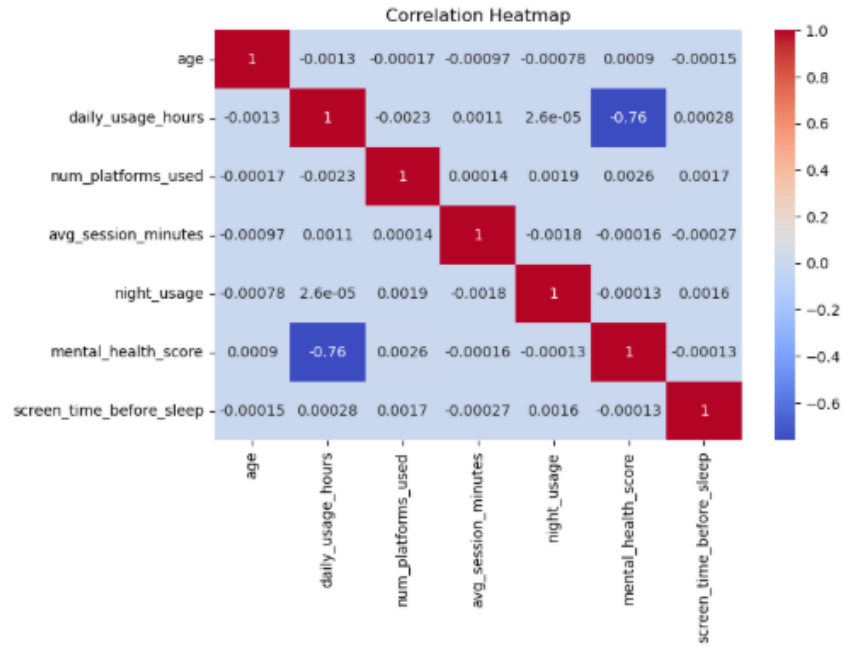
```
[25]: plt.figure(figsize=(8,5))
sns.countplot(x="addiction_level",data=df)
plt.title("Addiction level Distribution")
plt.xlabel("Addiction Level")
plt.ylabel("Counts")
plt.show()
```



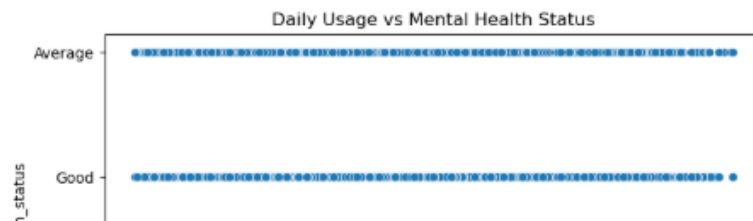
```
[26]: plt.figure(figsize=(8,5))
sns.countplot(x='age_group',data=df)
plt.title("Age Group Distribution")
plt.xlabel("Age Group")
```


Country

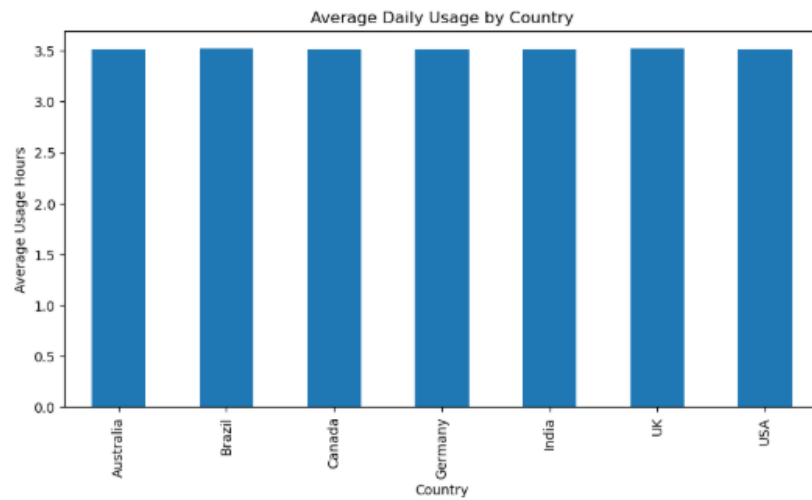
```
[28]: plt.figure(figsize=(8,5))
numeric_df=df.select_dtypes(include=np.number)
sns.heatmap(numeric_df.corr(),
            annot=True,
            cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```



```
[29]: plt.figure(figsize=(8,5))
sns.scatterplot(x='daily_usage_hours',y='mental_health_status',data=df)
plt.title("Daily Usage vs Mental Health Status")
plt.show()
```



```
plt.ylabel("Average Usage Hours")
plt.show()
```



```
[2]: pip install psycopg2 sqlalchemy
```

```
Requirement already satisfied: psycopg2 in d:\anaconda3\lib\site-packages (2.9.12)
Requirement already satisfied: sqlalchemy in d:\anaconda3\lib\site-packages (2.0.43)
Requirement already satisfied: greenlet>=1 in d:\anaconda3\lib\site-packages (from sqlalchemy) (3.2.4)
Requirement already satisfied: typing-extensions>=4.6.0 in c:\users\vijay kumar\appdata\roaming\python\python311\site-packages (from sqlalchemy) (4.1
5.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[3]: from sqlalchemy import create_engine
```

```
[23]: engine = create_engine('postgresql://postgres:postgres123@localhost:5432/genz_social_media')
```

```
[24]: df.to_sql('social_media_usage',engine,if_exists='replace',index=False)
```

```
[24]: 996
```

```
[34]: df = df[
      (df['mental_health_status'] != '#REF!') &
      (df['mental_health_status'] != '#SPILL!')
      ]
```

```
[35]: df['mental_health_status'].value_counts()
```

```
[35]: mental_health_status
Good      550981
Average   369526
Bad        75826
Poor        3663
Name: count, dtype: int64
```

```
[ ]:
```

7. Power BI Dashboard Development

Power BI was used to design and develop an interactive multi-page analytics dashboard with modern UI styling and advanced analytical visuals.

Dashboard Pages

1. Overview Dashboard

The Overview Dashboard provides a high-level summary of the social media dataset.

Key Features

- Total users
- Average daily usage
- Average mental health score
- Average session minutes
- Most used platform
- Platform popularity analysis
- Addiction level distribution
- Mental health status distribution

Visualizations Used

- KPI Cards
- Horizontal Bar Charts
- Doughnut Charts
- Column Charts
- Interactive Slicers



2. Behavior Analysis Dashboard

The Behavior Analysis page focuses on user engagement patterns and behavioral insights.

Key Features

- Average screen time
- Average platforms used
- Most active age group
- Platform engagement comparison
- Healthy usage target gauge
- Daily usage trend analysis
- Purpose of social media usage
- Screen time vs mental wellness analysis

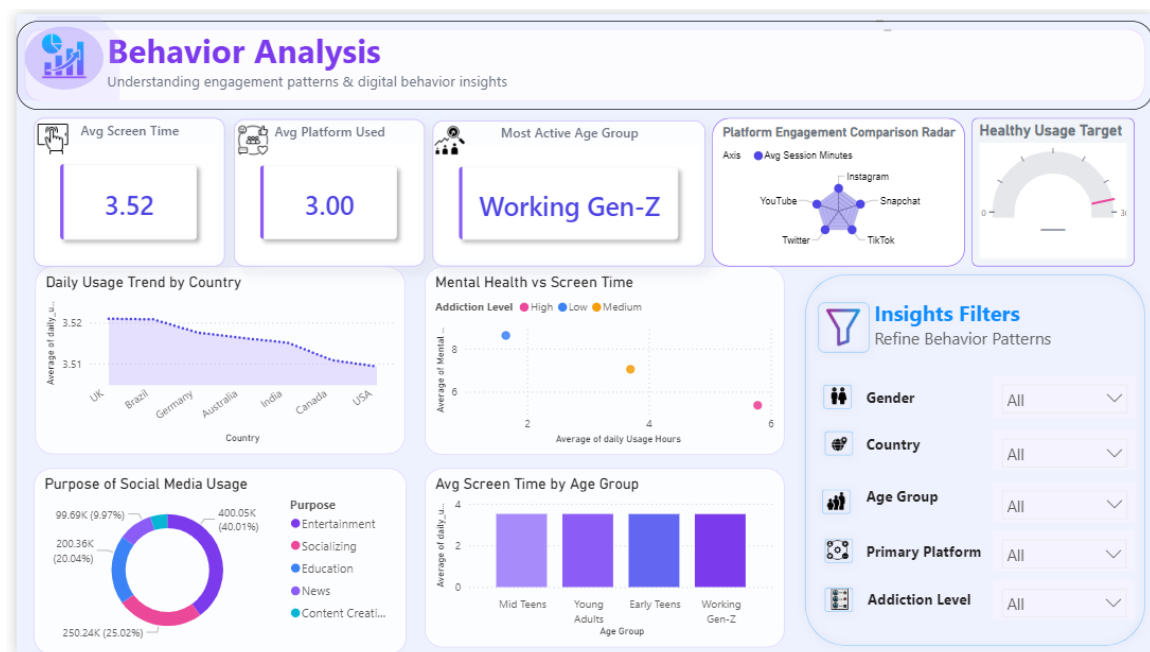
Visualizations Used

- Radar Chart
- Gauge Chart
- Scatter Plot

- Doughnut Chart
- Area Chart
- Column Chart

Insights Generated

- Young adults and working Gen-Z showed high engagement
- Increased screen time was associated with lower mental wellness
- Entertainment was the primary reason for social media usage
- Platform engagement varied significantly across user groups



3. Mental Health Insights Dashboard

The Mental Health Insights page focuses on emotional wellness and addiction-related analysis.

Key Features

- Average mental health score
- High addiction users
- Stress impact level

- Platform emotional impact
- Addiction level vs mental health
- Mental health distribution analysis
- Screen time vs mental wellness

Visualizations Used

- Radar Charts
- Scatter Charts
- Doughnut Charts
- Column Charts
- KPI Cards

Insights Generated

- Higher addiction levels showed lower mental health scores
- Excessive screen time negatively affected emotional wellness
- Social media engagement influenced stress and behavior patterns
- Mental health distribution varied across addiction categories



8. DAX Measures Used

Several DAX measures were created to improve dashboard interactivity and KPI reporting.

Important Measures

Average Screen Time

Avg Screen Time =

AVERAGE(public_social_media[daily_usage_hours])

Average Session Minutes

Avg Session Minutes =

AVERAGE(public_social_media[avg_session_minutes])

High Risk Users

High Risk Users =

CALCULATE(
COUNTROWS(public_social_media),
public_social_media[mental_health_score] <= 4
)

Stress Impact Level

Stress Impact Level =

VAR AvgUsage =

AVERAGE(public_social_media[daily_usage_hours])

VAR AvgMental =

AVERAGE(public_social_media[mental_health_score])

RETURN

ROUND((AvgUsage * 2) - (AvgMental / 2), 2)

These DAX measures helped generate meaningful analytical insights and KPI visualizations.

9. UI and Dashboard Design

The dashboard was designed with a modern analytics-focused theme using lavender, purple, blue, and pink color palettes.

Design Features

- Soft lavender analytics background
- Rounded KPI cards
- Custom icons and filter panels
- Interactive slicers
- Balanced chart layouts
- Modern enterprise-style dashboard structure
- Consistent visual hierarchy

The design focused on readability, analytical storytelling, and premium dashboard aesthetics.

10. Key Insights from the Project

The project generated several important insights:

- Instagram emerged as the most used platform among Gen-Z users
- Medium addiction users formed the largest user category
- Increased screen time showed negative correlation with mental health
- Young Adults and Working Gen-Z had higher engagement patterns
- Entertainment and social interaction were the major usage purposes
- Platform engagement directly influenced behavioral and emotional patterns

11. Skills Developed Through the Project

This project improved practical knowledge in:

- Data Cleaning
- SQL Querying
- Database Management

- Data Analysis
- Dashboard Design
- DAX Calculations
- Business Intelligence Reporting
- Data Storytelling
- UI/UX Dashboard Structuring
- Interactive Analytics Development

12. Business and Research Value

The dashboard can support:

- Digital wellness research
- Behavioral analytics studies
- Educational awareness programs
- Mental health analysis
- Social media engagement research
- Data-driven decision-making

This project demonstrates how business intelligence tools can be used to analyze human behavior and emotional wellness in the digital era.

13. Future Enhancements

Future improvements that can be implemented include:

- Real-time data integration
- AI-powered predictive analytics
- Sentiment analysis using NLP
- Mobile-responsive dashboard version
- User-level drill-through analytics
- Advanced forecasting and recommendation systems

14. Conclusion

The Gen-Z Social Media Usage Analysis Dashboard successfully transformed raw social media data into meaningful analytical insights using a complete end-to-end analytics workflow.

The project combined Excel preprocessing, Python analysis, PostgreSQL database management, and Power BI visualization to create a professional portfolio-ready business intelligence solution.

The dashboards effectively demonstrated behavioral patterns, engagement trends, addiction analysis, and mental health insights through interactive visual storytelling.

This project enhanced technical and analytical skills in data analytics, business intelligence, dashboard design, and modern reporting techniques while also providing valuable insights into Gen-Z digital behavior and emotional wellness.

15. Project Outcome

The project successfully achieved:

- Professional multi-page Power BI dashboards
- Advanced KPI and DAX implementation
- Interactive filtering and analytical visuals
- Modern dashboard UI/UX design
- End-to-end analytics workflow integration
- Portfolio-ready business intelligence project development

The project serves as a strong demonstration of practical skills in Excel, Python, PostgreSQL, Power BI, and data analytics.